Using genomic data to study population structure and local adaptation in *Boltonia decurrens*

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ABSTRACT

Boltonia decurrens (Torr. & Gray) Wood (Asteraceae) is a federally threatened perennial native to the Illinois River floodplain. Understanding its population structure and the extent to which it exhibits local adaptation to its environment is important for effective conservation in both wild and managed settings. Using whole genome sequencing of samples collected from throughout the range of the species, we examined the extent to which it hybridizes with co-occurring congeners, geographic patterns of genetic structure and diversity of *B. decurrens*, as well as whether it exhibits local adaptation to the environment. Three analytical methods (PCA, admixture, phylogenetic tree) identified 24 individuals from one site clustering with *Boltonia asteroides*, indicating that they were misidentified at the time of collection, but no individuals were admixed between the two species. Within *B. decurrens*, admixture analysis revealed strong genetic connectivity among geographically close populations and limited gene flow between distant populations, with a few genetically unique populations. We also detected isolation by distance (Mantel R = 0.428) when comparing F_{ST} values to geographic distance. However, isolation by environment was more strongly correlated (Mantel R = 0.494) with pairwise genetic differences. Our findings show no evidence of hybridization within our dataset and support the genetic distinctiveness of *B. decurrens* from closely related species. Genetic structure analyses indicate that gene flow occurs predominantly among neighboring populations and decreases with distance, consistent with isolation by distance shaped by the Illinois River. Furthermore, genetic differentiation correlated more strongly with environmental variables, particularly annual temperature range, than with geographic distance, suggesting local adaptation influences genetic variation. Our results underscore the importance of considering both geographic and environmental factors in conservation strategies to preserve the genetic diversity and adaptive potential of *Boltonia decurrens*.

1 | BACKGROUND

1.1 | Importance of studying population genetics

Ex situ conservation, defined by the Convention on Biological Diversity (2011) as “the conservation of components of biological diversity outside their natural habitats”, is a critical strategy for safeguarding threatened species. To maximize the effectiveness of ex situ collections, it is recommended that at least 95% of the genetic variation in populations be captured to maintain sufficient material for reintroduction (Akeroyd & Wyse Jackson, 1995). Genetic diversity is linked to individual-level fitness and supports adaptation to environmental change, highlighting its importance for conservation efforts (Reed & Frankham, 2003; Sgrò et al., 2011). However, resource constraints often limit the scope of collection efforts (Walls, 2018), making it unrealistic to collect individuals from the entire species range. Understanding genetic structure within and among populations can help overcome these limitations by informing conservation planning, identifying key populations for protection, and guiding practices that help preserve both species and their habitats (Mcmahon et al., 2014). Analyzing genetic structure can also reveal patterns of gene flow, habitat connectivity, seed dispersal, and other demographic processes that inform conservation efforts (Hamrick & Trapnell, 2011; Sork, 2015).

1.2 | Hybridization

Additionally, genetic analyses can detect instances of hybridization, which may influence conservation status and have both beneficial and detrimental effects on species fitness (Goulet et al., 2016). Hybridization between species poses a significant challenge to the long-term viability of threatened taxa. When a rare species hybridizes with a more common one, it can accelerate the decline of the rare species through reduced fertility, outbreeding depression, genetic swamping, and increased ecological competition from hybrids (Levin et al., 1996; Rhymer & Simberloff, 1996). However, interspecific hybridization can have advantageous effects, including hybrid vigor, transgressive segregation, adaptive introgression, and, in rare cases, hybrid speciation (Goulet et al., 2016). It is also important to understand if this process is occurring as analysis of population structure requires an understanding of the species boundaries, and if hybridization or misidentification occurs it can skew the results of the genetic analysis.

1.3 | Study Species

In this study we focused on *Boltonia decurrens* (Torr. and Gray) Wood (Asteraceae), which is a federally threatened perennial native to the Illinois River floodplain and near the confluence of the Illinois and Mississippi Rivers. This species is an early successional plant, relying on frequent flooding for its habitat and seed dispersal (Smith & Keevin, 1998). Due to *B. decurrens* need for regular disturbance, changes to the river-floodplain system have led to a decline in population (Smith et al., 2005). Both *B. decurrens* and the more widespread *B. asteroides* are endemic to this Illinois River floodplain, with the two being known to co-occur at some sites, and with some evidence of hybridization occurring (Dewoody et al., 2011).

1.4 | Study goals

The main goal of this study was to understand genetic diversity and structure in *Boltonia decurrens*, and how populations are locally adapted to their environments. Using whole-genome sequencing data, we analyzed the genetic structure of *B. decurrens* and compared it to *B. asteroides* to assess potential hybridization between these two species. We also examined the relationship between genetic and environmental variation to investigate genome-environment association. The specific goals were to 1) understand *Boltonia decurrens* evolutionary relationship with other species of *Boltonia* and if hybridization is occurring, 2) assess patterns of genetic diversity and genetic structure across the range of the species, and 3) investigate local adaptation of populations of *B. decurrens* to the local environment.

2 | MATERIALS AND METHODS

2.1 | Study Species, Sample Collection, and DNA Preparation

Samples of *Boltonia decurrens* were collected from multiple wild populations across its natural range to capture genetic diversity. Seeds were germinated and grown under controlled conditions. Leaf tissue was used for DNA extraction. Genomic DNA was extracted following standard protocols optimized for whole genome sequencing. Library preparation and sequencing were performed to generate high-quality whole genome data for downstream analyses.

2.3 | Variant filtering and quality control

Whole-genome variant calls for *Boltonia decurrens* were initially filtered using a custom hard-filtering pipeline built with bcftools v1.18 (Danecek et al., 2021). Filtering steps included the retention of only biallelic SNPs, masking of genotypes with GQ < 10 or depth < 3 or > 50, recalculation of INFO tags (AN, AC, AF, NS), and removal of variants with MAF < 0.01 or >10% missing data. Variants were further processed to ensure fixed diploid ploidy. Filtered VCFs were indexed and compressed for downstream use.

A second quality control step was performed using VCFtools v0.1.17 (Danecek et al., 2011), removing indels and further filtering for QUAL ≥ 30 and acceptable site-level depth metrics. The resulting dataset was used to assemble the phylogenetic tree (described in 2.4). For all other analysis, this dataset was converted to PLINK binary format (.bed or .psam). Individuals and loci with >10% missingness were removed using the --mind 0.1 and --geno 0.1 options in PLINK v1.9 (Chang et al., 2015). For neutral genetic analysis, linkage disequilibrium (LD) pruning was conducted using a 50-SNP window with a 10-SNP shift and an r^2 threshold of 0.1 (--indep-pairwise 50 10 0.1).

2.4 | Hybridization

PCA was conducted in PLINK v1.9 and ADMIXTURE was run using ADMIXTURE v1.3.0 (Alexander et al., 2009) on LD-pruned datasets with K = 2–20. A phylogenetic tree was constructed from the vcf files using IQTree (Nguyen et al., 2015).

2.4 | Genetic structure and diversity analysis

All potential misidentified individuals were removed, and PCA and ADMIXTURE were again conducted on this filtered LD-pruned datasets with K = 2–20. F_{ST} values were then calculated pairwise among defined populations (counties) using PLINK v2.0. To evaluate isolation by distance (IBD) and isolation by environment (IBE), we conducted Mantel tests comparing genetic distance matrices to either geographic distances (based on great-circle distances among population centroids) or environmental differences retrieved from WorldClim and ENVIREM datasets (Fick & Hijmans, 2017; Title & Bemmels, 2018). Mantel correlations were computed using the mantel() function from vegan (Oksanen et al., 2025), with results summarized across variables.

2.5 | Local Adaptation

PCAdapt (Luu et al., 2017) and LFMM (Frichot et al., 2013) were run on the non-LD-pruned datasets.

2.6 | Data visualization and spatial mapping

All figures were produced in R v4.5.0 (R Core Team, 2025). Data transformation and figure generation were conducted primarily using the tidyverse (Wickham et al., 2019). ggplot2 was used for PCA, ADMIXTURE barplots, and IBD/IBE scatterplots, with map-based graphics created using sf, ggspatial, scatterpie, and related packages. All shapefiles (including river networks) were sourced from publicly available databases. U.S. state and county boundaries were downloaded from Natural Earth and the U.S. Census Bureau via the tigris R package. Coordinates were projected using the EPSG:3857 projection, and manual offsets were applied to overlapping pie charts. F_{ST} versus distance and environmental difference plots were annotated with Mantel r and p-values and fit with linear regression lines. Summary results were visualized as bar charts displaying Mantel *r* values across variables.

3 | RESULTS

3.1 | Hybridization

To look for evidence of hybridization, we ran a PCA, admixture analysis, and phylogenetic tree. In our PCA, the first and second principal component accounted for 21.44% and 8.75% of the variation within the data respectively (Figure 1). The PCA grouped the data into four distinct clusters, and when coloring each individual based on their sample group, 24 individuals from the Cass sample clustered with all outgroup species from *B. asteroides* and *B. diffusa*, we subsequently labeled them as misID as shown in Figure 1. Admixture analysis was performed for K values ranging from 1 to 20. The cross-validation error was highest at K = 1 (0.32966) and lowest at K = 8 (0.3027), which was therefore selected for downstream analysis (Figure S1, Table S1). Our admixture plot reveals a shared genetic ancestry between all outgroup individuals and the potentially misidentified samples. Minor proportions of this ancestry are also observed in the Saint Clair group and in two individuals from the Schuyler group (<0.074). Additionally, several outgroup individuals exhibit low levels of non-outgroup ancestry (0.047-0.16), with one individual (477) showing a notably high proportion (0.53) of non-outgroup ancestry (Figure 2, S2, Table S2). Our phylogenetic analysis places all 24 potentially misidentified individuals within a single monophyletic clade, along with one *B. asteroides* individual (478). Of the outgroup individuals we ran admixture analysis on, the group of 25 individuals (misID and 478) is positioned between a set of six individuals that branch off earlier and another set of five that appear more closely related to the remaining *B. decurrens* samples. The rest of *B. decurrens* forms a distinct monophyletic group, which splits from a single *B. asteroides* individual (477) (Figure 2, S3).

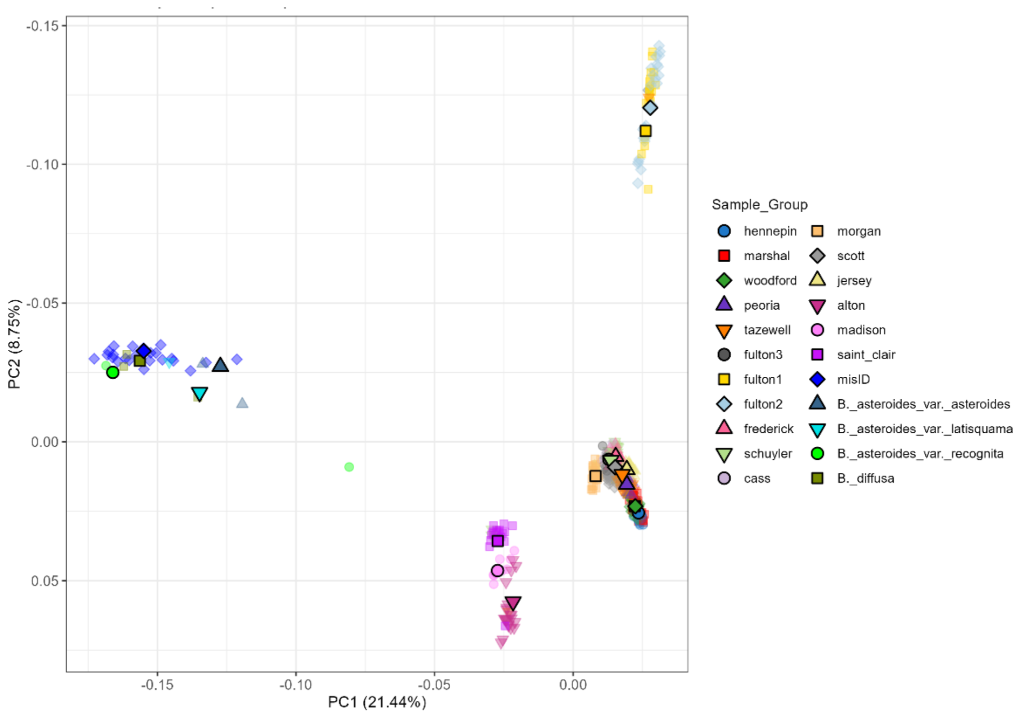

Figure 1. PCA with genetically similar clusters from dataset including outgroups.

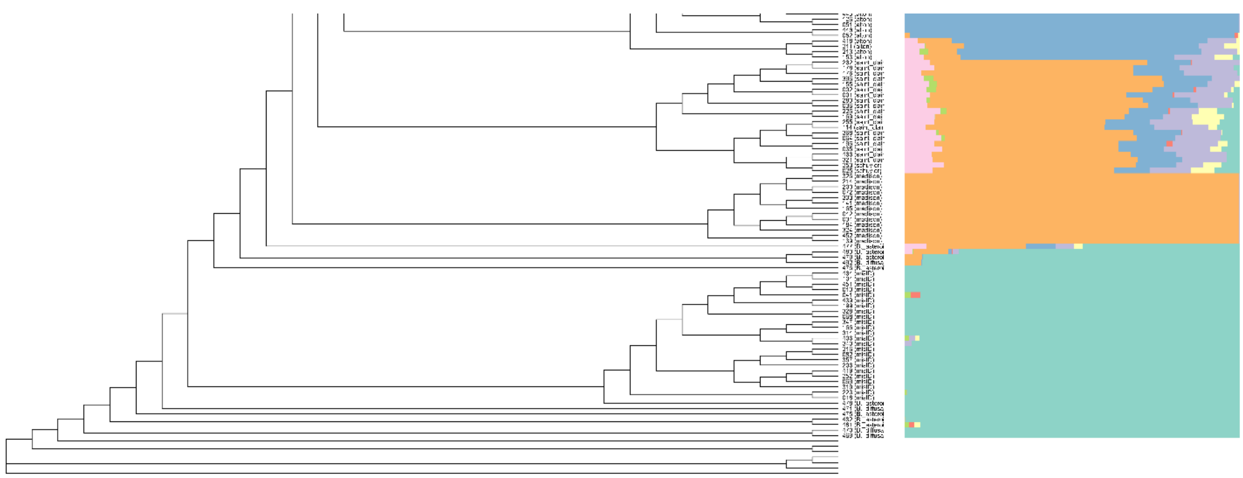

Figure 2. Phylogenetic tree zoomed in to see outgroups and closely related B. decurrens. Non-labeled nodes are outgroups that admixture was not ran on.

3.2 | Genetic Structure

To analyze genetic structure within *Boltonia decurrens*, we used PCA, admixture analysis, and F_{st} with isolation by distance (IBD) on a filtered dataset with the 24 misID individuals from Cass removed. In our PCA, the first and second principal component accounted for 2.64% and 2.43% of the variation within the data respectively (Figure 3C). The PCA revealed three broadly defined clusters: one consisting of the Fulton 1 and 2 sample groups; a second comprising the Saint Clair, Madison, and Alton groups; and a third including the remaining sample groups. While some individuals from one sampling group clustered with another, the broader clustering pattern was largely maintained. Excluding Fulton 1 and 2, the distribution of samples largely followed a latitudinal gradient, though some deviations from this pattern were observed, the most notable being Jersey.

Admixture analysis was performed on this filtered data with K values ranging from 1 to 20. The cross-validation error was highest at K = 1 (0.40959) and lowest at K = 8 (0.37375), which was therefore selected for downstream analysis (Figure S4, Table S3). The admixture analysis revealed a general pattern of gene flow among geographically proximate populations, with limited admixture occurring across long distances (Figure 3A,B). Overall, few populations appeared genetically distinct, with widespread admixture observed across most individuals, many of whom exhibited contributions from all eight ancestral populations. Notably, the Fulton 1 and 2 groups formed a distinct genetic cluster, showing minimal evidence of shared ancestry with other populations within these groups. Comparing F_{ST}, which is a measure of genetic difference between a pair of populations, to distance shows a significant correlation between the two, with greater correlation shown when looking at river distance ($R = 0.428$) as opposed to geographic distance ($R = 0.382$).

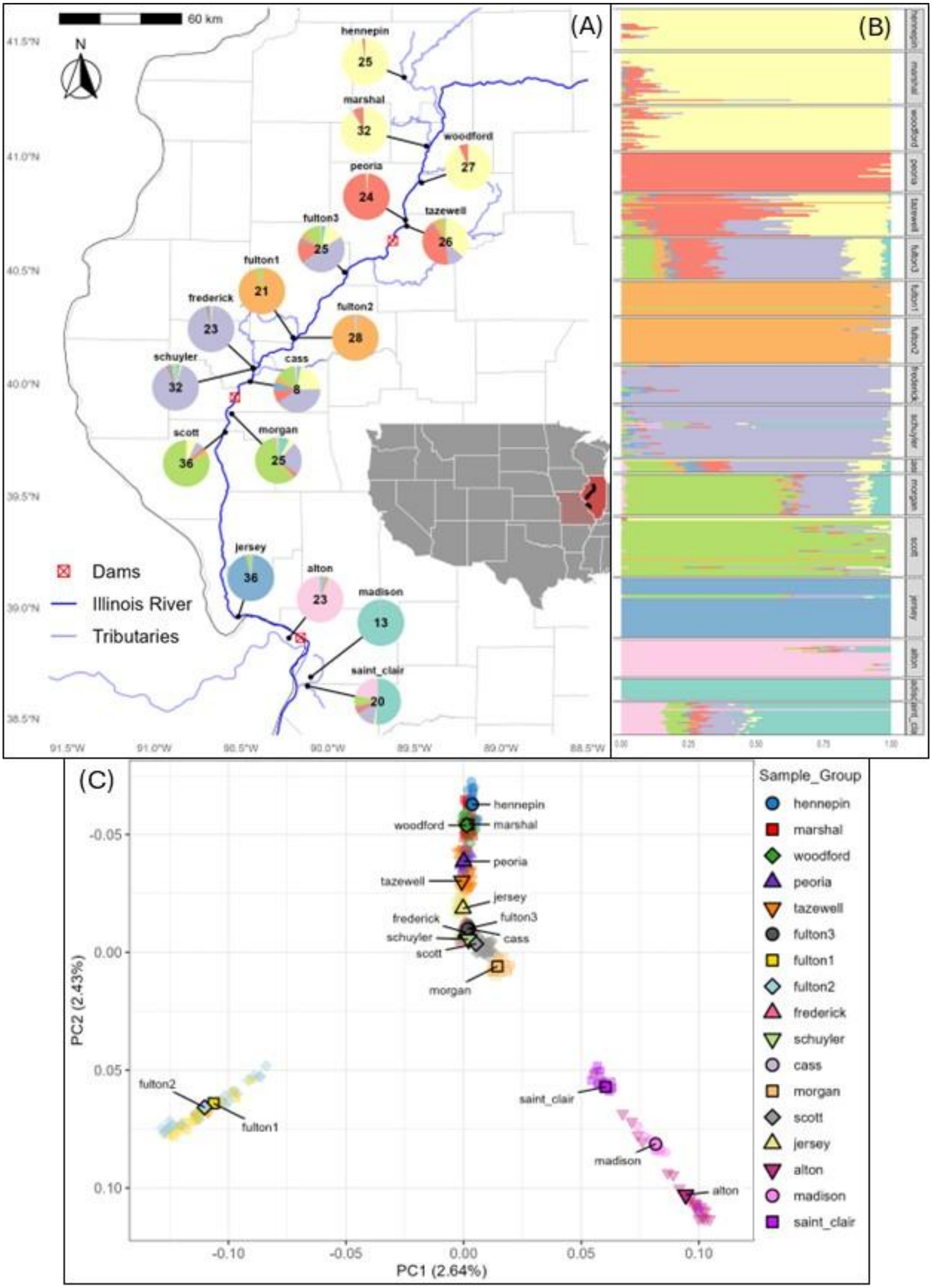

Figure 3. (A) Map with admixture results displayed as pie charts for each population along with sample sizes. (B) ADMIXTURE Analysis plot showing ancestral populations for true B. decurrens. Each line is the genetic composition of an individual, which is grouped by geographic

population. Number of populations defined by K = 8 (lowest CV error). (C) PCA with genetically similar clusters from filtered dataset without misID from Cass.

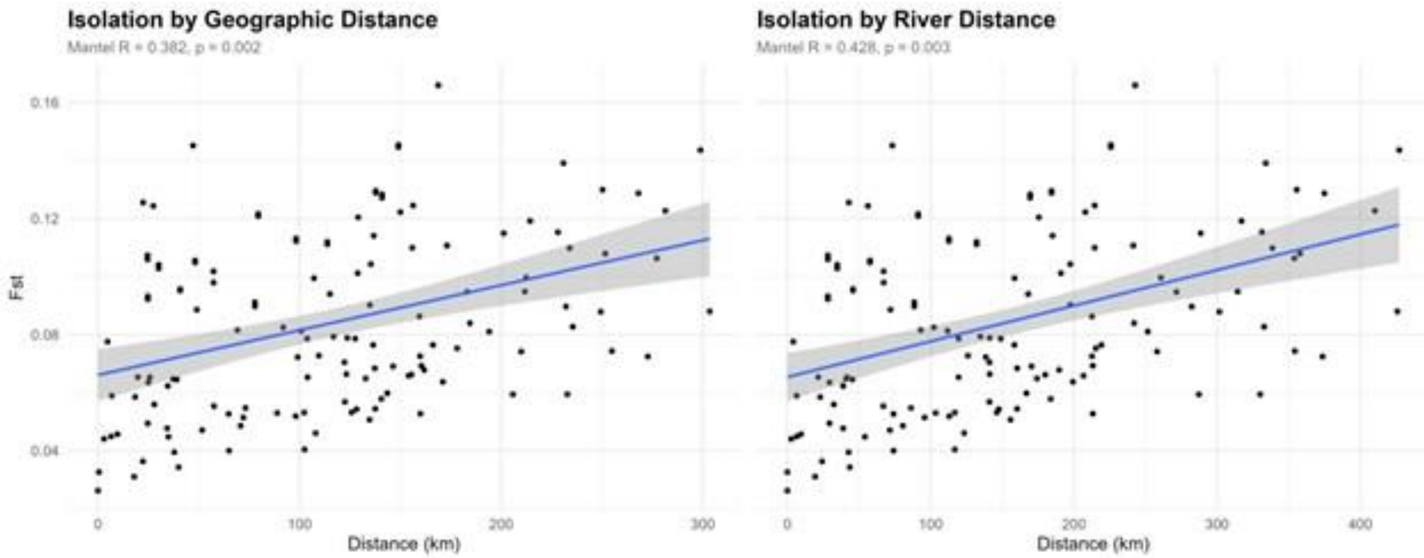

Figure 4. F_{ST}, a measure of genetic difference between pairs of populations, plotted against geographic distance and distance along river between pairs of populations. Mantel R, a measure of correlation, and p-value displayed.

3.3 | Local Adaptation

To evaluate potential environmental influences on genetic differentiation, we conducted a Mantel test comparing pairwise F_{ST} values to differences in annual temperature range (Figure 5). This analysis yielded a Mantel R of 0.494 ($p = 0.004$), which was higher than the Mantel R for river distance (0.428, $p = 0.003$), previously reported (Figure 4, 5).

We also ran genome-environment association analyses using PCAdapt (K = 8) and LFMM (K = 2), with LFMM incorporating annual temperature range as the environmental variable. Both analyses identified loci with elevated $-\log_{10}$ (p-values), with a shared peak in chromosome 1. LFMM also identified several peaks across chromosomes 2, 7, and 8 that were not detected in PCAdapt which are highlighted in green (Figure 6).

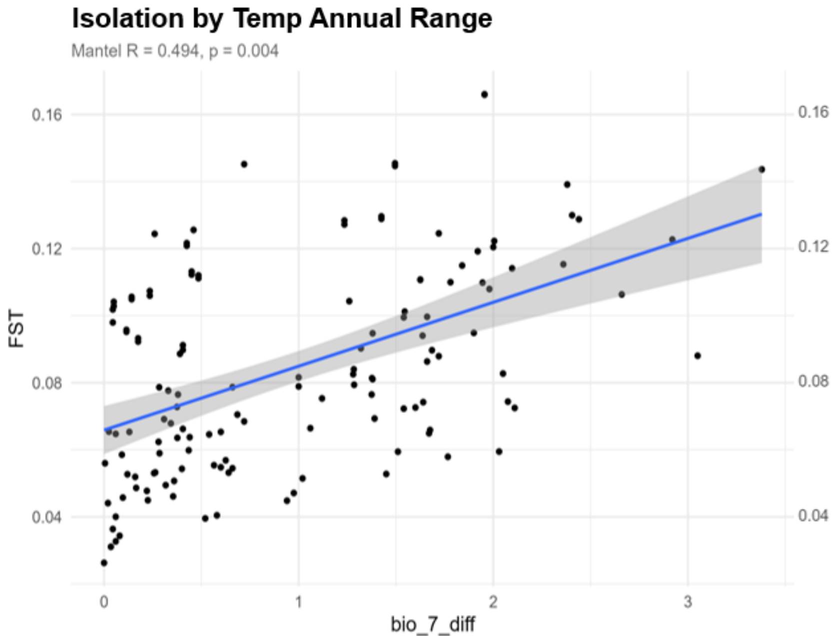

Figure 5. F_{ST} plotted against difference in annual temp ranges between pairs of populations. Mantel R, a measure of correlation, and p-value displayed.

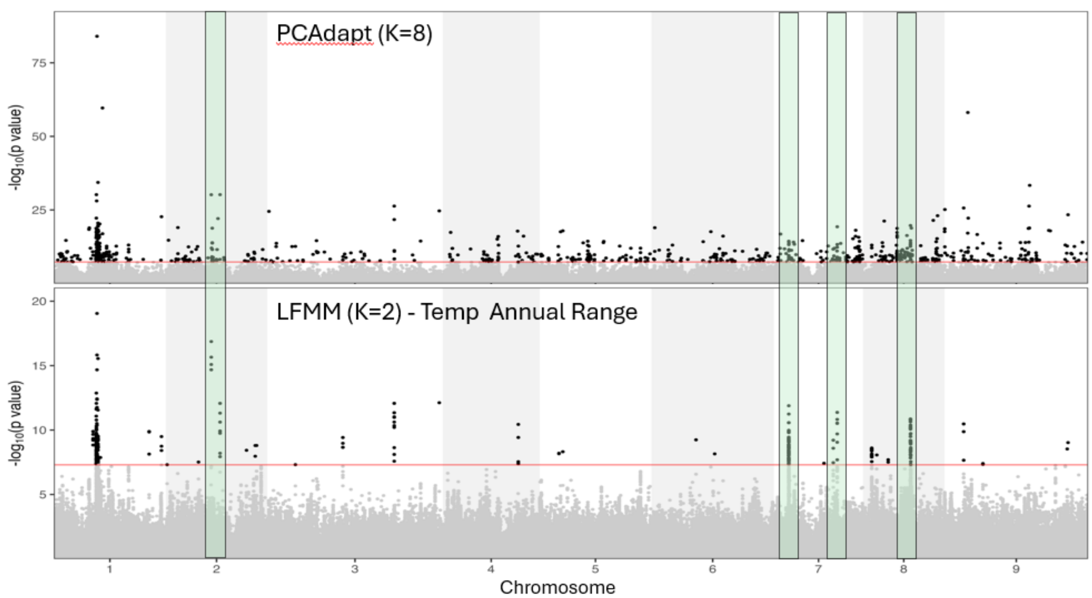

Figure 6. PCAdapt and LFMM showing loci with allele frequencies highly correlated with population structure (PCAdapt) or annual temp range (LFMM). Peaks present in LFMM but not in PCAdapt highlighted green

4 | DISCUSSION

4.1 | Hybridization

We identified 24 individuals from the Cass population that clustered with outgroup species *Boltonia asteroides* and *B. diffusa* in both PCA and phylogenetic analyses. These individuals likely represent misidentifications rather than hybrids, as they form a distinct clade with no evidence of intermediate ancestry between *B. decurrens* and *B. asteroides*. Previous research by DeWoody et al. (2011) found limited hybridization between

B. decurrens and *B. asteroides* at two sympatric sites. Morphological classification identified four individuals with intermediate assignment probabilities, and genetic analysis only detected low frequencies of F_2 and backcross individuals, but no F_1 hybrids were found. In contrast, in our dataset we observed no intermediate genotypes beyond the 24 misidentified individuals, and no evidence of hybrid individuals among verified *B. decurrens* samples.

4.2 | Genetic Structure

Our analyses of population structure revealed high levels of admixture among close populations and low admixture between distant populations indicating that gene flow is primarily occurring between relatively nearby populations and over the course of the entire species range gene flow is limited. PCA and admixture results showed a north-to-south gradient in genetic variation, with certain populations such as Fulton 1 and 2 forming distinct genetic clusters. This pattern is further supported by the significant relationship between pairwise F_{ST} and river distance, indicating isolation by distance (IBD) as an important factor shaping genetic differentiation in *B. decurrens*. The presence of widespread admixture in most populations also suggests that gene flow occurs frequently between closer populations, possibly facilitated by river dynamics.

4.3 | Local Adaptation

We found some evidence of local adaptation in *B. decurrens* populations. The Mantel test showed that genetic differentiation was more strongly correlated with differences in annual temperature range (Mantel $R = 0.494$) than with river distance ($R = 0.428$), indicating that some adaptation to temperature may be occuring. Genome-environment association analysis using LFMM identified multiple loci significantly associated with annual temperature range, with notable peaks on chromosomes 2, 7, and 8. PCAdapt detected a more limited set of outlier loci, with overlap between the two models in chromosome 1. The unique peaks found in the LFMM but not in PCAdapt suggest that there may be specific loci that are more strictly associated with adaptation to temperature over just the general adaptation found in population structure.

4.4 | Implications for Conservation

The absence of widespread hybridization within *B. decurrens* supports its status as a genetically distinct group which is good for the survival of this threatened species. The observed genetic structure, characterized by localized gene flow indicates that conservation strategies should aim to preserve a broad range of populations across the species' geographic range. This approach would help maintain existing genetic diversity and local adaptations, particularly considering environmental variation. Further work is needed to identify and validate candidate genes involved in adaptation, which could guide future conservation and restoration efforts in the context of our changing climate and with the threats that humans pose on the survival of this species.

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REFERENCES

Akeroyd, J., & Wyse Jackson, P. (1995). *A handbook for botanic gardens on the reintroduction of plants to the wild*.

Alexander, D. H., Novembre, J., & Lange, K. (2009). Fast model-based estimation of ancestry in unrelated individuals. *Genome Research*, *19*(9), 1655–1664. https://doi.org/10.1101/GR.094052.109,

Chang, C. C., Chow, C. C., Tellier, L. C. A. M., Vattikuti, S., Purcell, S. M., & Lee, J. J. (2015). Second-generation PLINK: Rising to the challenge of larger and richer datasets. *GigaScience*, *4*(1). https://doi.org/10.1186/S13742-015-0047-8,

Danecek, P., Auton, A., Abecasis, G., Albers, C. A., Banks, E., DePristo, M. A., Handsaker, R. E., Lunter, G., Marth, G. T., Sherry, S. T., McVean, G., & Durbin, R. (2011). The variant call format and VCFtools. *Bioinformatics*, *27*(15), 2156–2158. https://doi.org/10.1093/BIOINFORMATICS/BTR330

Danecek, P., Bonfield, J. K., Liddle, J., Marshall, J., Ohan, V., Pollard, M. O., Whitwham, A., Keane, T., McCarthy, S. A., & Davies, R. M. (2021). Twelve years of SAMtools and BCFtools. *GigaScience*, *10*(2). https://doi.org/10.1093/GIGASCIENCE/GIAB008,

Dewoody, J., Nason, J. D., & Smith, M. (2011). Hybridization between the threatened herb Boltonia decurrens (Asteraceae) and its widespread congener, B. asteroides. *Botany*, *89*(3), 191–201. https://doi.org/10.1139/B11-005

Fick, S. E., & Hijmans, R. J. (2017). WorldClim 2: new 1-km spatial resolution climate surfaces for global land areas. *International Journal of Climatology*, *37*(12), 4302–4315. https://doi.org/10.1002/JOC.5086

Frichot, E., Schoville, S. D., Bouchard, G., & François, O. (2013). Testing for Associations between Loci and Environmental Gradients Using Latent Factor Mixed Models. *Molecular Biology and Evolution*, *30*(7), 1687–1699. https://doi.org/10.1093/MOLBEV/MST063

Goulet, B. E., Roda, F., & Hopkins, R. (2016). Hybridization in Plants: Old Ideas, New Techniques. *Plant Physiology*, *173*(1), 65. https://doi.org/10.1104/PP.16.01340

Hamrick, J. L., & Trapnell, D. W. (2011). Using population genetic analyses to understand seed dispersal patterns. *Acta Oecologica*, *37*(6), 641–649. https://doi.org/10.1016/J.ACTAO.2011.05.008

Levin, D. A., Francisco-Ortega, J., & Jansen, R. K. (1996). Hybridization and the extinction of rare plant species. *Conservation Biology*, *10*(1), 10–16. https://doi.org/10.1046/J.1523-1739.1996.10010010.X;PAGEGROUP:STRING:PUBLICATION

Luu, K., Bazin, E., & Blum, M. G. B. (2017). pcadapt: an R package to perform genome scans for selection based on principal component analysis. *Molecular Ecology Resources*, *17*(1), 67–77. https://doi.org/10.1111/1755-0998.12592

Mcmahon, B. J., Teeling, E. C., & Höglund, J. (2014). How and why should we implement genomics into conservation? *Evolutionary Applications*, *7*(9), 999. https://doi.org/10.1111/EVA.12193

Nguyen, L. T., Schmidt, H. A., Von Haeseler, A., & Minh, B. Q. (2015). IQ-TREE: A Fast and Effective Stochastic Algorithm for Estimating Maximum-Likelihood Phylogenies. *Molecular Biology and Evolution*, *32*(1), 268–274. https://doi.org/10.1093/MOLBEV/MSU300

Oksanen, J., Simpson, G. L., Blanchet, F. G., Kindt, R., Legendre, P., Minchin, P. R., O'Hara, R. B., Solymos, P., Stevens, M. H. H., Szoecs, E., Wagner, H., Barbour, M., Bedward, M., Bolker, B., Borcard, D., Borman, T., Carvalho, G., Chirico, M., De Caceres, M., … Weedon, J. (2025). Community Ecology Package [R package vegan version 2.7-1]. *CRAN: Contributed Packages*. https://doi.org/10.32614/CRAN.PACKAGE.VEGAN

R Core Team. (2025). *R: A Language and Environment for Statistical Computing* (4.5.0). R Foundation for Statistical Computing.

Reed, D. H., & Frankham, R. (2003). Correlation between fitness and genetic diversity. *Conservation Biology*, *17*(1), 230–237. https://doi.org/10.1046/J.1523-1739.2003.01236.X;WEBSITE:WEBSITE:CONBIO

Rhymer, J. M., & Simberloff, D. (1996). Extinction by hybridization and introgression. *Annual Review of Ecology and Systematics*, *27*(Volume 27, 1996), 83–109. https://doi.org/10.1146/ANNUREV.ECOLSYS.27.1.83/CITE/REFWORKS

Secretariat of the Convention on Biological Diversity. (2011). *Convention on Biological Diversity: Text and Annexes*. Secretariat of the Convention on Biological Diversity. https://www.cbd.int/doc/legal/cbd-en.pdf

Sgrò, C. M., Lowe, A. J., & Hoffmann, A. A. (2011). Building evolutionary resilience for conserving biodiversity under climate change. *Evolutionary Applications*, *4*(2), 326–337. https://doi.org/10.1111/J.1752-4571.2010.00157.X;CTYPE:STRING:JOURNAL

Smith, M., Caswell, H., & Mettler-Cherry, P. (2005). Stochastic flood and precipitation regimes and the population dynamics of a threatened floodplain plant. *Ecological Applications*, *15*(3), 1036–1052. https://doi.org/10.1890/04-0434

Smith, M., & Keevin, T. M. (1998). Achene morphology, production and germination, and potential for water dispersal in Boltonia decurrens (decurrent false aster), a threatened floodplain species. *Rhodora*, *100*(901), 69–81. https://www.jstor.org/stable/23313269?seq=1&cid=pdf-

Sork, V. L. (2015). Gene flow and natural selection shape spatial patterns of genes in tree populations: implications for evolutionary processes and applications. *Evolutionary Applications*, *9*(1), 291. https://doi.org/10.1111/EVA.12316

Title, P. O., & Bemmels, J. B. (2018). ENVIREM: an expanded set of bioclimatic and topographic variables increases flexibility and improves performance of ecological niche modeling. *Ecography*, *41*(2), 291–307. https://doi.org/10.1111/ECOG.02880

Walls, S. C. (2018). Coping with constraints: Achieving effective conservation with limited resources. *Frontiers in Ecology and Evolution*, *6*(MAR). https://doi.org/10.3389/fevo.2018.00024

Wickham, H., Averick, M., Bryan, J., Chang, W., D', L., Mcgowan, A., François, R., Grolemund, G., Hayes, A., Henry, L., Hester, J., Kuhn, M., Lin Pedersen, T., Miller, E., Bache, S. M., Müller, K., Ooms, J., Robinson, D., Seidel, D. P., … Yutani, H. (2019). Welcome to the Tidyverse. *Journal of Open Source Software*, *4*(43), 1686. https://doi.org/10.21105/JOSS.01686